

Implementation & Evaluation Plan

CV-Based Town Planning Risk System

What We Are Building, Which Datasets We Check Against, and Exactly How Success Is Measured

Aligned against our base paper: Mohamadiazar et al. (2024), Journal of Hydrology

1. Purpose of This Document

This document is written to leave no open question about two things: exactly how we will build this project step by step, and exactly what we will check our results against to prove they are correct — not just what metric name we will report, but which real, named dataset supplies the ground truth for every single number in our results section.

Our evaluation approach deliberately follows the same structure as our base paper (Mohamadiazar et al., 2024, Journal of Hydrology) — the same category of metrics (accuracy, precision, recall, F1, AUC, and a ground-truth validation index) — so our results can be discussed side by side with a published, peer-reviewed benchmark, even though our target output and use case are different.

2. Actual Implementation Plan

2.1 Stage 1 — Data Collection & Alignment

All datasets are pulled for one fixed bounding box (our chosen town) so every data layer covers the exact same ground:

Layer	Source	Resolution	Role
Satellite imagery (2 years)	Sentinel-2 Level-2A, via AWS Earth Search STAC	10 m, bands B02/B03/B04/B08	Input to the segmentation model; used for change detection
Elevation	Copernicus Global DEM (GLO-90/GLO-30)	30-90 m	Identifies low-lying, flood-prone zones
Water bodies / rivers	OpenStreetMap hydrography	Vector	Used to compute distance-to-water for every location
Building footprints (labels)	OSM + Microsoft Global ML Building Footprints + Google Open Buildings v3	Vector polygons	Ground truth for training and testing the segmentation model

Elevation and satellite data are resampled onto one shared pixel grid (resolution alignment) before any further processing, so that a given pixel means the same physical patch of ground across every layer.

2.2 Stage 2 - Building / Land-Cover Segmentation Model

- Model: DeepLabV3+ (CNN-based semantic segmentation), pretrained on ImageNet, fine-tuned on our local data
- Classes: Building / Vegetation / Other
- Input patch size: 256x256 pixels (same convention as the base paper)
- Training labels: building footprint polygons (OSM + Microsoft + Google), rasterized into pixel masks matching each image patch
- Train/test split: 70% training, 30% testing (same ratio as the base paper) - the 30% test set is never shown to the model during training and is the exact set used to compute all Level 1 metrics below
- Data augmentation: horizontal/vertical flipping and rotation, to increase training diversity (same approach as the base paper)
- Optimizer: Adam; Loss function: Dice loss (standard for segmentation class-imbalance); Activation: ReLU in hidden layers, sigmoid/softmax at output

2.3 Stage 3 - Multi-Year Change Detection

The trained model is run separately on the earlier-year and later-year image sets. The two output masks are compared pixel-by-pixel:

```
new_construction = (mask_later == BUILDING) AND (mask_earlier != BUILDING)
green_loss        = (mask_earlier == VEGETATION) AND (mask_later != VEGETATION)
```

2.4 Stage 4 - Risk Scoring (Our Core Novel Contribution)

```
HIGH RISK    = new construction + low elevation + near water
MEDIUM RISK = new construction + (low elevation OR near water)
LOW RISK     = new construction + neither hazard factor present
```

"Low elevation" = bottom 25-30th percentile of elevation values in the study area. "Near water" = within a fixed buffer distance (e.g. 350 m) of a mapped water body. Both thresholds are stated explicitly as heuristic starting points, not calibrated flood-engineering values, and are tested in the ablation study in Section 7.

2.5 Stage 5 - Dashboard & Citizen Lookup

The final risk map is served through a planner-facing web dashboard and a citizen-facing address lookup tool that returns a plain-language risk summary for any searched location within our covered town boundary.

3. Evaluation Overview - Three Independent Levels

Our project has three components that can each succeed or fail independently: the segmentation model, the change-detection step, and the final risk score. We evaluate each one separately, against its own specific ground truth, rather than judging the whole system by one single number.

Level	What is being checked	Ground truth used	Type of check
1. Segmentation model	Does the model correctly identify buildings/vegetation in a satellite image?	Held-out 30% of OSM + Microsoft + Google building-footprint labels	Automatic, pixel-by-pixel comparison
2. Change detection	Is flagged "new construction" actually new construction?	Manual visual cross-check against raw imagery and Google Earth historical imagery (time-slider)	Manual sample verification
3. Risk scoring	Do our flagged high-risk zones actually correspond to real flood-affected areas?	Dartmouth Flood Observatory (DFO) 2018 Kerala flood-extent GIS data (real satellite-derived flood polygons)	Automatic spatial overlay comparison

4. Level 1 - Segmentation Model Evaluation (Full Detail)

4.1 What We Compare Against

Ground truth: the 30% of building-footprint polygons (from OSM, Microsoft Global ML Building Footprints, and Google Open Buildings) that are held back from training. These are converted into pixel masks at the same resolution as our satellite image patches. Every prediction the model makes on the test set is compared pixel-by-pixel against this held-out mask - this is a genuine, standard, defensible baseline, not something we are approximating.

4.2 Metrics, Formulas, and What They Mean

Metric	Formula	What it tells us	Base paper's reported result
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall % of correctly classified pixels	97.05% (train) / 93.73% (test)

Metric	Formula	What it tells us	Base paper's reported result
Precision	$TP / (TP + FP)$	Of pixels we called "building", how many really were	97.03% (train) / 94.07% (test)
Recall	$TP / (TP + FN)$	Of all real building pixels, how many we correctly found	88.62% (train) / 86.00% (test)
F1 Score	$2 \times (Precision \times Recall) / (Precision + Recall)$	Balanced combination of precision and recall	92.49% (train) / 89.85% (test)
AUC-ROC	Area under the ROC curve	How well the model separates building vs. non-building overall (1.0 = perfect, 0.5 = random guessing)	0.93
IoU (Intersection over Union)	Overlap area / Union area of predicted vs. true building shape	Stricter shape-overlap metric - not just "pixel correct" but "boundary correct"	Not reported by base paper - our addition for rigor

TP = True Positive (correctly predicted building), TN = True Negative (correctly predicted non-building), FP = False Positive (predicted building but wasn't), FN = False Negative (missed a real building) - all computed directly from the confusion matrix between our prediction and the held-out label mask.

4.3 Our Target Range

We aim for results in the same broad band as the base paper (roughly 85-95% across accuracy/precision/recall/F1), while explicitly noting in our report that Indian urban imagery (denser, more irregular building layouts) is a harder segmentation problem than the base paper's water-detection task on US suburban terrain - so a modest gap below their numbers is expected and will be discussed honestly, not hidden.

5. Level 2 - Change Detection Evaluation (Full Detail)

5.1 Why There Is No Ready-Made Dataset Here

No public dataset lists "every new building constructed in this specific town between these two exact years." This is a genuine, disclosed gap - not unique to our project; the base paper's own flood-inventory step also relied on a semi-automatic method (Otsu thresholding) rather than an independent labeled dataset.

5.2 What We Compare Against Instead

- A random sample of locations flagged as "new construction" (e.g. 50 zones) is manually cross-checked against Google Earth's historical imagery time-slider, which lets us visually scrub between years for the same coordinates and confirm whether real construction occurred
- Where available, we cross-reference against any local news coverage or municipal records of specific development projects in the study area, as a secondary sanity check

5.3 What We Report

A manual verification rate: "Out of N sampled flagged zones, X were visually confirmed as genuine new construction." This is reported as a supporting sanity-check figure, not a headline accuracy percentage, and is explicitly labeled in our report as a manual, sample-based check rather than an exhaustive automated evaluation.

6. Level 3 - Risk Score Validation (Full Detail)

6.1 The Real Ground-Truth Dataset We Use

This is the equivalent of the base paper's Ground Truth Index step (where they compared predictions against 311 flood complaint records and Twitter reports). We use a genuine, publicly available, satellite-derived flood-extent dataset:

Property	Detail
Dataset	Dartmouth Flood Observatory (DFO), Global Active Archive of Large Flood Events - 2018 India/Kerala flood record (Event #4663)
What it contains	GIS flood-extent shapefiles/GeoTIFFs showing the maximum observed flood inundation area, derived from real Sentinel-1 SAR and Landsat 8 satellite data
Coverage relevance	Directly covers Kerala's 2018 floods, the same event context as our Periyar-basin study area (Aluva / North Kochi)
Access	Free, public, provided by the University of Colorado Boulder - usable for non-commercial academic work under a Creative Commons Attribution-NonCommercial-ShareAlike license (attribution required)
Source link	floodobservatory.colorado.edu (DFO Event #4663, 2018 India)

This gives us something the base paper effectively also had: an independent, real record of where flooding actually occurred, that we did not generate ourselves and cannot be accused of tuning our results to match after the fact.

6.2 How the Comparison Works

1. Overlay our generated High-Risk zone polygons on the same map as the DFO flood-extent polygon for our study area
2. For each of our flagged High-Risk zones, check whether it falls inside or very near the DFO-documented flood-extent boundary
3. Calculate our Risk Validation Rate: (Number of our High-Risk zones that overlap real documented flood extent) divided by (Total number of our High-Risk zones flagged), x 100%

6.3 Formula

Risk Validation Rate = (High-Risk zones overlapping DFO flood extent / Total High-Risk zones flagged) x 100%

-- Direct structural equivalent of the base paper's Ground Truth Index:

Ground Truth Index = (Observed flood events matching prediction / Total observed flood events) x 100%

6.4 Honest Limitation We Disclose

The DFO dataset records maximum flood extent for the entire 2018 event at a regional scale - it is not a block-by-block, building-level flood damage survey the way the base paper's 311-complaint database was. We state this openly: our Risk Validation Rate measures whether our flagged construction-risk zones fall within a real, independently documented flood-affected region, which is a meaningful and legitimate check, but at a coarser spatial precision than the base paper's point-level validation. This is disclosed as a known scope difference, not presented as equivalent precision.

7. Ablation Study - Testing Which Risk Factors Actually Matter

The base paper ran an 8-case ablation study (their Table 4), testing different combinations of input data to see which factors most improved flood prediction. We run the equivalent test for our risk-scoring logic:

Case	Inputs Used	Purpose
1	New construction only	Baseline - risk signal from growth alone, no hazard factors
2	New construction + elevation only	Isolate the contribution of elevation alone
3	New construction + water proximity only	Isolate the contribution of water proximity alone
4	New construction + elevation + water proximity (full model)	Our complete risk model - expected best performance

Each case's flagged High-Risk zones are separately checked against the DFO flood-extent ground truth (Section 6), producing a Risk Validation Rate per case. We expect - and will test rather than assume - that Case 4 (combined) outperforms Cases 1-3, mirroring the base paper's own finding that the model trained with all input data performed best.

8. Full List of Comparison Parameters

This is the complete set of parameters we report in our results section, what each is measured against, and its direct counterpart in the base paper.

#	Parameter	Measured On	Compared Against (Ground Truth)	Base Paper Equivalent
1	Accuracy	Segmentation model	Held-out building-footprint labels (OSM/MS/Google)	97.05% / 93.73%
2	Precision	Segmentation model	Held-out building-footprint labels	97.03% / 94.07%
3	Recall	Segmentation model	Held-out building-footprint labels	88.62% / 86.00%
4	F1 Score	Segmentation model	Held-out building-footprint labels	92.49% / 89.85%
5	AUC-ROC	Segmentation model	Held-out building-footprint labels	0.93
6	IoU	Segmentation model	Held-out building-footprint labels	Not reported (our addition)
7	Manual verification rate	Change detection	Google Earth historical imagery (manual sample check)	Not directly comparable - new step for our project
8	Risk Validation Rate	Risk scoring (final output)	Dartmouth Flood Observatory 2018 Kerala flood-extent GIS data	Ground Truth Index (84.05%)
9	Ablation comparison (4 cases)	Risk scoring	Same DFO flood-extent data, applied per case	Input-combination Table (8 cases, base paper)

9. When Each Evaluation Happens

- Level 1 (segmentation metrics) - evaluated immediately after model fine-tuning, before change detection begins
- Level 2 (manual change-detection check) - evaluated once new-construction flags are generated for our two comparison years
- Level 3 (Risk Validation Rate + ablation study) - evaluated last, once the full risk-scoring pipeline is complete; this forms the core results section of our final report

10. One-Paragraph Summary (For Quick Reference)

"We evaluate our system at three independent levels, each against a specific, named ground truth: our segmentation model is tested against a held-out 30% split of real building-footprint labels from OSM, Microsoft, and Google; our change-detection step is manually sample-verified against Google Earth historical imagery; and our final risk score is validated against the Dartmouth Flood Observatory's real, satellite-derived 2018 Kerala flood-extent data - producing a Risk Validation Rate that is the direct structural equivalent of our base paper's Ground Truth Index. Every metric we report has a specific, real dataset behind it - nothing is evaluated without a stated baseline."

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